Geophys. J. R. astr. Soc. (1967) **12**, 469–471.

Palaeomagnetism and Age of Rocks from Easter Island and Juan Fernandez

J. Booker, E. C. Bullard and R. L. Grasty

(Received 1967 January 3)

Downloaded from https://academic.oup.com/gji/article/12/5/469/747578 by Curtin University Library user on 26 May 2026

In 1966 the expedition, '*Carousel*' of the Scripps Institution of Oceanography called for three days at Easter Island and for a few hours at Juan Fernandez. The opportunity was taken to collect rock samples of known orientation for the determination of their ages and directions of magnetization. Owing to the shortness of the available time, the investigation was less thorough than was desirable. A Brunton compass was used to orient the samples.

Easter Island

Fourteen samples of basalt were collected from seven locations on the south and west sides of the island. An attempt was made to obtain the oldest exposed rocks by collecting near the base of the sea cliffs. The results of the determinations of the direction of magnetization on thirty-one specimens cut from ten of the samples are shown in Fig. 1. The results from two samples (eight specimens) were widely

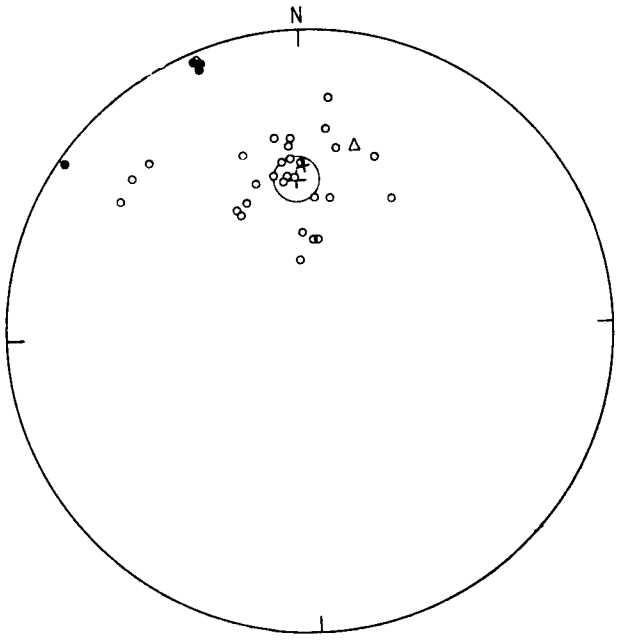


FIG. 1. Directions of magnetization of rocks from Easter Island. ○ upward (normal) dips, ● downward dips, + observed mean direction ignoring the eight observations in the upper left of the figure; the circle shows the area in which the true mean has a 95% chance of lying. △ present field, × axial dipole field.

scattered and discrepant from the rest, these were omitted in determining the mean direction. The mean direction for the remaining twenty-three specimens is declination 3° W, dip −48° (i.e., normal for the southern hemisphere). The radius, α, of the circle in which the true mean lies with 95%, probability is 5·5° and Fisher's measure of dispersal, κ, is 31. All these quantities were calculated as described by Irving (1964), an equal weight being given to each specimen.

The direction of the field at Easter Island due to an axial dipole would be declination 0°, dip −45°. This direction lies 4° from the observed mean direction of magnetization and is within the 95% confidence circle. The axial diple field is closer to the observed mean than is the present field (declination 15·5° E, dip −37·5°). The intensities of magnetization lay in the range 0·002 to 0·01 e.m.u. with a mean of 0·005 and the ratio of ' permanent ' to induced magnetization in the range 0·4 to 5·0 with a mean of 3·2.

An attempt was made to determine the ages of six specimens by the potassium-argon method without separation of minerals. All the results were less than one million years and did not differ significantly from zero, the greatest was $0{\cdot}72 \pm 0{\cdot}43$ m.y.

The results therefore lead to the expected conclusion for an island near the axis of a mid-ocean ridge. The exposed rocks of Easter Island are less than one million years old, and are magnetized in a direction within a few degrees of that of the field of an axial dipole.

Juan Fernandez

Twelve samples of basalt were collected from West Bay and Cumberland Bay on the north side of the larger and more easterly island of Más Atierra and two from the western island, Más Afuera. From these, thirty-three specimens were cut for the measurement of the direction of magnetization. The results are shown in Fig. 2. The dip is downwards (i.e. reversed for the southern hemisphere) for all the samples, but the directions are so scattered in declination that no useful mean can be taken.

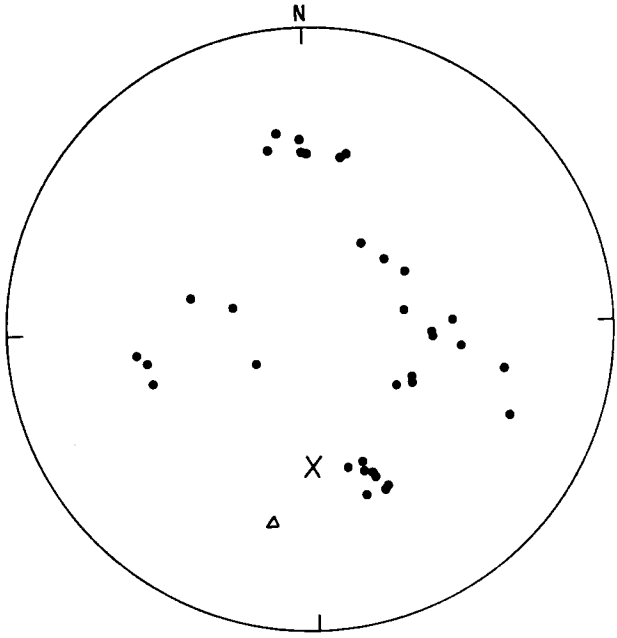


FIG. 2. Directions of magentization of rocks from Juan Fernandez. All show downward (reversed) dips. △ present field, × axial dipole field (both upward).

Downloaded from https://academic.oup.com/gji/article/12/5/469/747578 by Curtin University Library user on 26 May 2026

The reason for this scatter is unknown. Partial demagnetization in an a.c. field of up to 800 gauss did not reduce the scatter. The intensities of magnetization ranged from 0·0002 to 0·008 e.m.u. with a mean of 0·0025, the ratio of ' permanent ' to induced magnetization in the range 0·2 to 4·5 with a mean of 1·4. These values are lower than those for the Easter Island rocks and suggest possible instability of magnetization.

The ages of four samples were determined by the potassium–argon method without separation of minerals. The results are given in Table 1.

Table 1

Sample No.	Island	K_20 %	Vol. radiogenic argon per g (10^{-5}mm^3)	Atmospheric contamination	Age (m.y.)
880	Más Afuera	0·915	3·84	86·7	1·3±0·3
880	Más Afuera	0·915	2·58	93·9	0·85±0·3
956	Más Afuera	0·963	3·23	93·0	1·0±0·1
881	Más Atierra	0·315	3·71	92·1	3·5±0·8
882	Más Atierra	0·880	9·06	82·9	3·1±0·9

$\lambda e = 0{\cdot}584 \times 10^{-10}\ y^{-1}$ $\lambda\beta = 4{\cdot}72 \times 10^{-10}\ y^{-1}$

The specimens were collected from a small area and must not be assumed to represent the whole range of ages present; the islands may well have a long history of vulcanism. It does, however, seem likely that Juan Fernandez is older than Easter Island. This was to be expected since the islands are deeply dissected in contrast to the almost unweathered appearance of the ash cones of Easter Island. The difference is consistent with Wilson's (1963) view that islands on the crest of a mid-ocean ridge are younger than those on either side, the difference is, however, less than might be expected from Wilson's examples for an island 1800 km from the axis of the ridge.

Further investigations of the palaeomagnetism and ages of Juan Fernandez is desirable. To collect a representative set of specimens would be a considerable task owing to the inaccessibility of the interior of the islands.

Department of Geodesy and Geophysics,
Madingley Rise,
Madingley Road,
Cambridge.

University of California,
San Diego.

1967 *January.*

References

Irving, E., 1964. *Palaeomagnetism*, Chapter 4. Wiley, New York.
Wilson, J. T., 1963. *Nature, Lond.*, **198**, 925–929.

Downloaded from https://academic.oup.com/gji/article/12/5/469/747578 by Curtin University Library user on 26 May 2026